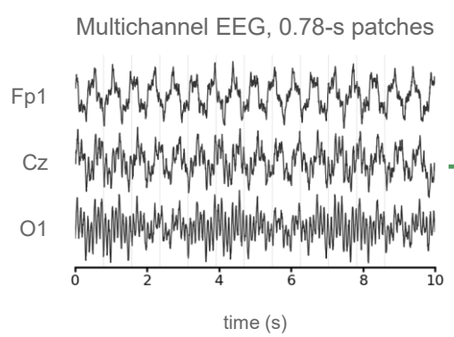
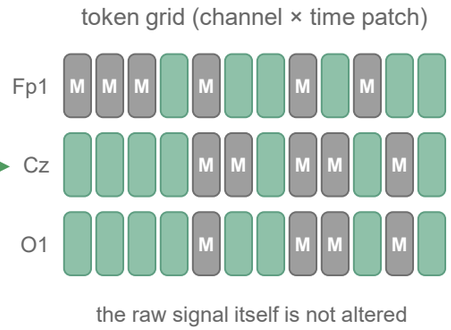
(a) Pretraining

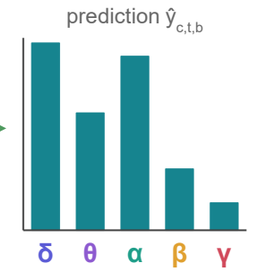


Patchify + linear embedding

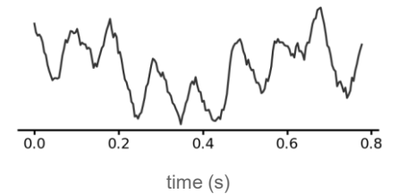


Transformer encoder
criss-cross: channel / time attention

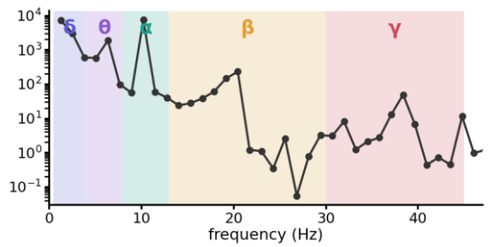
Band-power head



same, uncorrupted signal



FFT

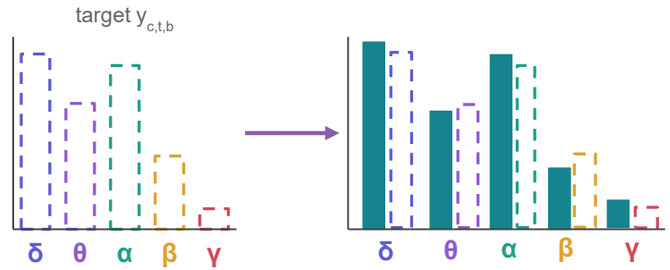


δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–45 Hz

Parameter-free target

1. Periodogram $|X_k|^2$
2. Band power $P_{c,t,b} = (2/P^2)\ \Sigma_{f_k \in B_b}\ |X_k|^2$
3. Log $y_{c,t,b} = \log(P_{c,t,b} + \varepsilon)$

masked patches only



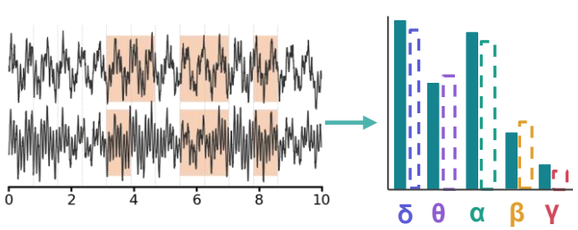
$$\mathcal{L} = 1/(5|\mathcal{M}|)\ \Sigma_{(c,t) \in \mathcal{M}}\ \Sigma_b\ (\hat{y}_{c,t,b} - y_{c,t,b})^2$$

no reconstruction term, no regularizer

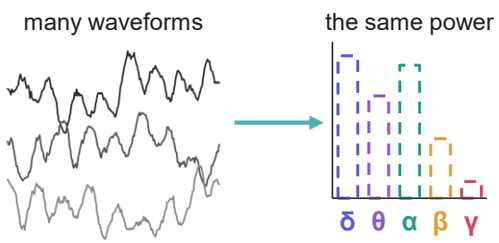
trainable · predicted band power · target band power · hidden patch (panel b)

(b) What the target asks the encoder to learn

ours — Predict band power

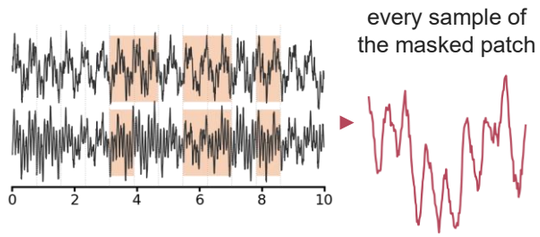


infer the hidden patch's power from its context

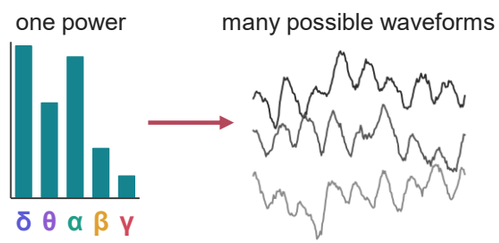


phase is unpredictable and irrelevant: learn the spectral state

alternative — Reconstruct the raw waveform



infer the hidden patch's waveform from its context



capacity spent on phase and noise